

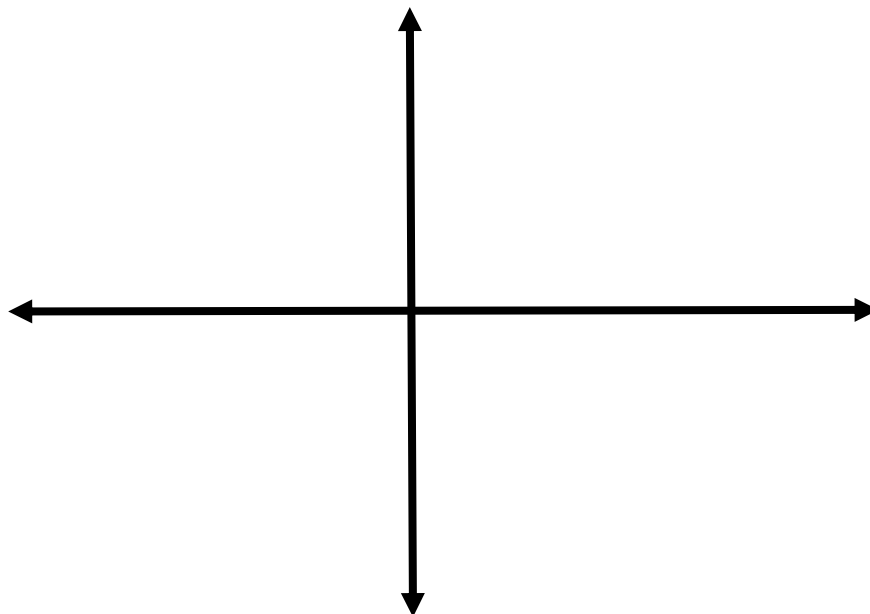
Vertical, Horizontal and Oblique Asymptotes

Vertical Asymptotes (V. A.)

We can identify Vertical Asymptotes (V. A.) by finding x values that make the denominator equals 0.

If $x = a$ is a V. A., calculate $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$. This limit will either be $+\infty$ or $-\infty$. To decide which test values near the V. A.

Example 1: Sketch the graph of $f(x) = \frac{2(x^2 + 1)}{(x - 1)(x + 2)}$ near its vertical asymptotes.



Horizontal Asymptotes (H. A.)

- The location of the horizontal asymptotes can be found by determining $\lim_{x \rightarrow \infty} f(x)$ or $\lim_{x \rightarrow -\infty} f(x)$.
- Determine the value of $f(x)$ for both small and large values of x .
 - If the value is greater than the value of the H. A., then the function is approaching the H. A. from above.
 - If the value is less than the value of the H. A., then the function is approaching the H. A. from below.

Example 2: Determine the H. A. of $f(x) = \frac{3-2x}{x-3}$. Also determine where the function is above and below the H. A.

Example 3: Determine if $f(x) = \frac{3x^2 + 2x - 1}{x^2 - 4}$ crosses the H. A. If so, determine the point of intersection.

Example 4: Determine if $f(x) = \frac{x^3 - 13x + 12}{x^2 - 3x - 10}$ crosses the oblique asymptote. If so, determine the point of intersection.

Example 5: Determine if $f(x) = \frac{2x^3 - x^2 + 3}{x^2}$ crosses the oblique asymptote. If so, determine the point of intersection.